

Part 2: Quantitative data analysis

Quantitative Session of the 2026 HUSOE ELPS Methods and Data Institute

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0.0.1 The Big Picture

Prior to jumping into more details about data analysis, and how we might be more critical in our analyses, it will be important to gather a “big picture” view of the field. This big picture view will provide us with a high-level description of the various topics we will cover in the course.

0.0.1.1 Population parameter

A number which summarizes the entire group.

0.0.1.2 Sample statistic

A single number that summarizes a subset of data, or the sample.

0.0.1.3 Confidence interval (CI)

- Range of likely or plausible values for a population parameter.
 - Based on a sample and statistics from that sample.
 - *Margin of error* represents the number of standard deviations on a statistic.
-

0.0.1.4 Hypothesis testing

Statistical procedure used to test an existing claim about a population.

- Test is based on data; most ideally data collected via probability sampling.
 - H_0 - Null hypothesis: If data supports the claim, fail to reject H_0
 - H_a - Alternative hypothesis: If data does not support claim, reject H_0
-

0.0.1.5 Analysis of variance (ANOVA)

Comparing means of more than two populations.

F-statistic is a ratio that is used to compare variability between sets.

0.0.1.5.1 Multiple comparison procedures

Set of statistical tests that compare means to each other.

- Examples include Turkey's test, Least significant difference (LSD), pairwise t-test
 - These tests are only conducted if you analysis of variance identifies differences
-

0.0.1.6 Interaction effects

Interaction effects are relevant to statistical models that use two or more variables.

0.0.1.7 Correlation

Measures the strength and direction of a linear relationship between two variables.

0.0.1.8 Linear regression

Helps make predictions for one variable based on the values of another.

There are multiple types of regression:

- Simple linear regression
 - Multiple linear regression
 - Logistic regression
 - Non-linear regression
-

0.0.1.9 Chi-square test

When using correlation and regression analyses, one core assumption is that the variables are quantitative in nature.

We use a chi-square test to study categorical variables.

0.1 Discussion: What types of statistical tests will you be using in your dissertation?

0.2 Case Study: HBCUs and Educational Geography

In this section, we analyze data on Historically Black Colleges and Universities (HBCUs) to explore patterns in location, institutional type, and founding history.

0.2.1 Step 1: Load Packages

```
#install.packages("tidyverse")
#install.packages("lubridate")
library(tidyverse)
library(lubridate)
```

We'll start with a guided example using data on Historically Black Colleges and Universities.

0.2.2 Step 2: Load the data

```
library(tidyverse)
hbcu <- read_csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/main/data/hbcu.csv")

# View the first few rows
head(hbcu)
```

```
# A tibble: 6 x 7
  name                city      state founded   lat   lon type
  <chr>              <chr>    <chr>   <dbl> <dbl> <dbl> <chr>
1 Alabama A&M University Normal    AL      1875  34.8 -86.6 Public, 4 Year
2 Alabama State University Montgomery AL      1867  32.4 -86.3 Public, 4 Year
3 Albany State University Albany     GA      1903  31.6 -84.2 Public, 4 Year
4 Alcorn State University Lorman    MS      1871  31.9 -91.1 Public, 4 Year
5 Allen University     Columbia SC      1870  34.0 -81.0 Private, 4 Year
6 American Baptist College Nashville TN       1924  36.2 -86.8 Private, 4 Year
```

```
# Check structure
glimpse(hbcu)
```

Rows: 102

Columns: 7

```
$ name      <chr> "Alabama A&M University", "Alabama State University", "Albany ~
$ city      <chr> "Normal", "Montgomery", "Albany", "Lorman", "Columbia", "Nashv~
$ state     <chr> "AL", "AL", "GA", "MS", "SC", "TN", "AR", "SC", "NC", "FL", "A~
$ founded   <dbl> 1875, 1867, 1903, 1871, 1870, 1924, 1884, 1870, 1873, 1904, 19~
$ lat       <dbl> 34.7834, 32.3643, 31.5785, 31.8769, 34.0298, 36.1659, 34.7465,~
$ lon       <dbl> -86.5683, -86.2952, -84.1543, -91.1458, -81.0115, -86.7844, -9~
$ type      <chr> "Public, 4 Year", "Public, 4 Year", "Public, 4 Year", "Public,~
```

```
# Summary statistics
summary(hbcu)
```

name	city	state	founded
Length:102	Length:102	Length:102	Min. :1837
Class :character	Class :character	Class :character	1st Qu.:1870
Mode :character	Mode :character	Mode :character	Median :1886
			Mean :1895
			3rd Qu.:1905
			Max. :1988

lat	lon	type
Min. :18.34	Min. :-98.50	Length:102
1st Qu.:32.48	1st Qu.: -90.13	Class :character
Median :34.02	Median :-84.64	Mode :character
Mean :34.31	Mean :-85.13	
3rd Qu.:36.17	3rd Qu.: -80.78	
Max. :39.93	Max. :-64.96	

0.2.3 Step 3: Clean the data

```
hbcu <- hbcu %>%
  mutate(
    founded = as.numeric(founded),
    type = as.factor(type),
    state = as.factor(state)
  )
```

0.2.4 Step 4: Basic Exploration

How many institutions are in the data set?

```
nrow(hbcu)
```

```
[1] 102
```

What is the distribution by HBCU type?

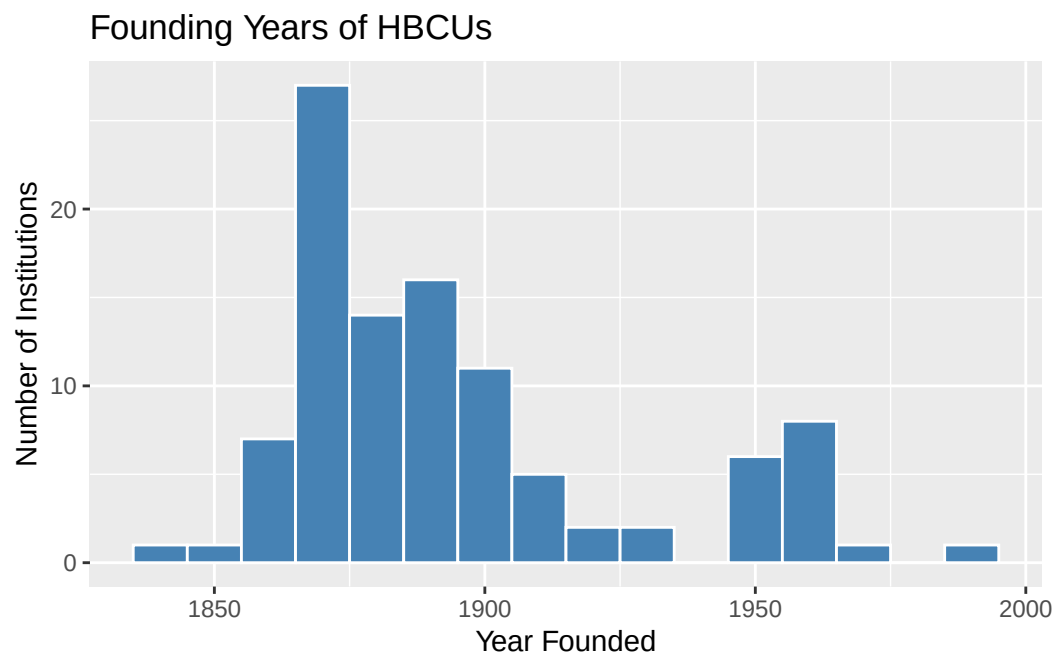
```
hbcu %>%
  count(type) %>%
  arrange(desc(n))
```

```
# A tibble: 5 x 2
  type                n
  <fct>              <int>
1 Private, 4 Year     45
2 Public, 4 Year       40
3 Public, 2 Year       11
4 Private, Specialized 4
5 Private, 2 Year       2
```

0.2.5 Step 5: Historical Analysis

When were the HBCUs founded?

```
hbcu %>%
  ggplot(aes(x = founded)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  labs(
    title = "Founding Years of HBCUs",
    x = "Year Founded",
    y = "Number of Institutions"
  )
```



0.2.6 Step 6: Geographic Distribution

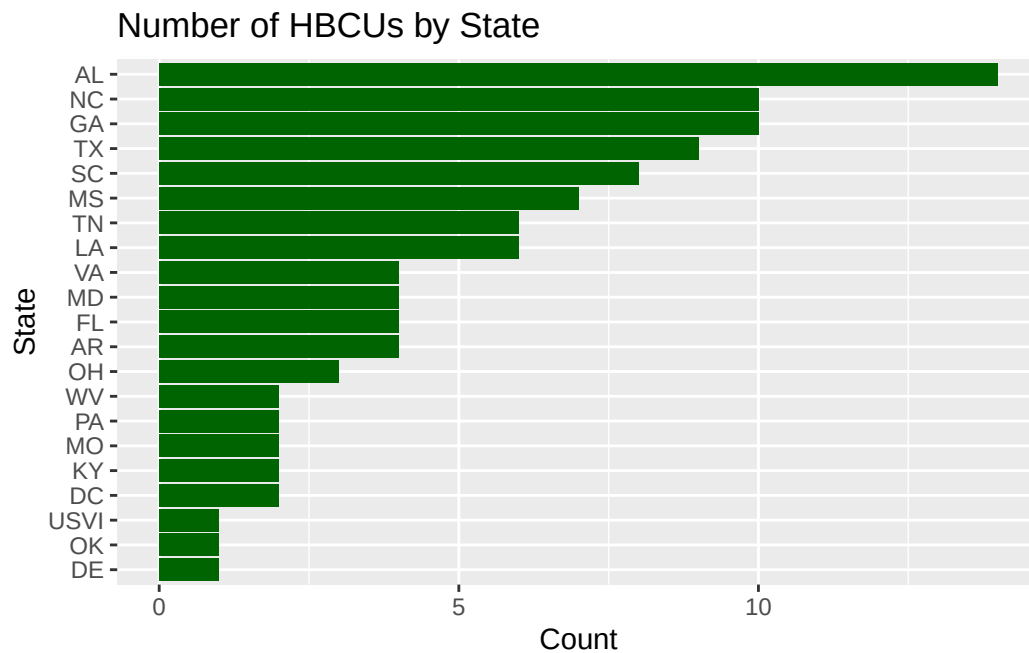
HBCUs by state

```
hbcu %>%  
  count(state, sort = TRUE)
```

```
# A tibble: 21 x 2  
  state      n  
  <fct> <int>  
1 AL      14  
2 GA      10  
3 NC      10  
4 TX       9  
5 SC       8  
6 MS       7  
7 LA       6  
8 TN       6  
9 AR       4  
10 FL      4  
# i 11 more rows
```

Basic visualization of HBCUs by state

```
hbcu %>%  
  count(state) %>%  
  ggplot(aes(x = reorder(state, n), y = n)) +  
  geom_col(fill = "darkgreen") +  
  coord_flip() +  
  labs(  
    title = "Number of HBCUs by State",  
    x = "State",  
    y = "Count"  
  )
```

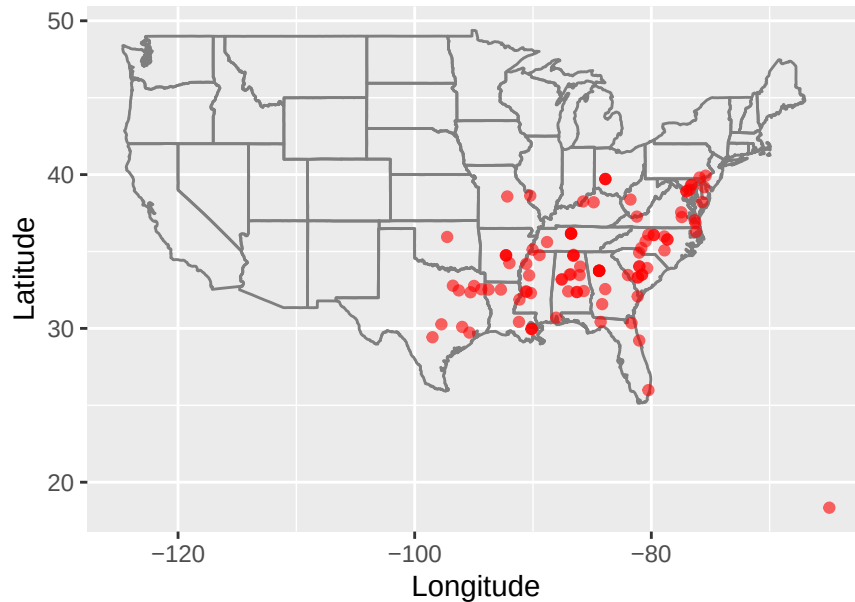


0.2.7 Step 7: Mapping

```
#install.packages("maps", repos = "http://cran.us.r-project.org")
library(maps)
```

```
hbcu %>%
  ggplot(aes(x = lon, y = lat)) +
  borders("state") +
  geom_point(color = "red", alpha = 0.6) +
  coord_fixed(1.3) +
  labs(
    title = "Geographic Distribution of HBCUs in the U.S.",
    x = "Longitude",
    y = "Latitude"
  )
```


Geographic Distribution of HBCUs in the U.S.



0.3 Case Study: Data on State School Districts

0.3.1 Step 1: Load the libraries

```
library(dplyr)
library(readr)
library(stringr)
library(janitor)
```

0.3.2 Step 2: Load the data

```
district <- read_csv(
  "https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/main/data/state-district-data.csv"
  show_col_types = FALSE
)

# district <- read_csv("../data/state-district-data.csv")

str(district)
```

```
spc_tbl_ [57 x 36] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ State Name
```

```
: chr [1:57] "ALABAMA" "
```

\$ State Abbr	: chr [1:57]	"AL" "AK" "A
\$ ANSI/FIPS State Code [State] Latest available year	: chr [1:57]	"01" "02" "
\$ Total Number Operational School Districts [District] 2023-24	: num [1:57]	156 54 1 68
\$ Total Number Operational Schools [Public School] 2023-24	: num [1:57]	1524 494 29
\$ Total Number Operational Charter Schools [Public School] 2023-24	: chr [1:57]	"18" "31" "
\$ Prekindergarten Students [State] 2023-24	: chr [1:57]	"21924" "33
" "20810" ...		
\$ Kindergarten Students [State] 2023-24	: chr [1:57]	"55881" "92
" "72320" ...		
\$ Grades 1-8 Students [State] 2023-24	: chr [1:57]	"447317" "7
" "654843" ...		
\$ Grades 9-12 Students [State] 2023-24	: chr [1:57]	"223528" "3
" "369124" ...		
\$ Grade 13 Students [State] 2023-24	: chr [1:57]	"†" "†" "†
\$ Ungraded Students [State] 2023-24	: chr [1:57]	"†" "†" "†
\$ American Indian/Alaska Native Students [State] 2023-24	: chr [1:57]	"5997" "282
" "46878" ...		
\$ Asian or Asian/Pacific Islander Students [State] 2023-24	: chr [1:57]	"11016" "64
" "35975" ...		
\$ Hispanic Students [State] 2023-24	: chr [1:57]	"84196" "10
" "539087" ...		
\$ Black or African American Students [State] 2023-24	: chr [1:57]	"236304" "2
" "64938" ...		
\$ White Students [State] 2023-24	: chr [1:57]	"381550" "6
" "378320" ...		
\$ Nat. Hawaiian or Other Pacific Isl. Students [State] 2023-24	: chr [1:57]	"831" "4148
" "4017" ...		
\$ Two or More Races Students [State] 2023-24	: chr [1:57]	"28756" "17
" "48415" ...		
\$ Total Enrollment (Exclude AE) for SY 2014-15 onward [State] 2023-24	: chr [1:57]	"748650" "1
" "1117630" ...		
\$ Full-Time Equivalent (FTE) Teachers [State] 2023-24	: chr [1:57]	"42857.92" "
" "49260.47" ...		
\$ Pupil/Teacher Ratio [State] 2023-24	: chr [1:57]	"17.47" "18
" "22.69" ...		
\$ State Name [State] 2023-24	: chr [1:57]	"ALABAMA" "A
\$ Total Number of Districts with Enrollment [District] 2023-24	: num [1:57]	150 54 0 66
\$ Total Number of School Districts [District] 2023-24	: num [1:57]	159 54 1 72
\$ Total Number of Public Schools [Public School] 2023-24	: num [1:57]	1527 498 29
\$ Male Students [State] 2023-24	: chr [1:57]	"384500" "6
\$ Female Students [State] 2023-24	: chr [1:57]	"364150" "6
\$ Black or African American - male [State] 2023-24	: chr [1:57]	"120605" "1
" "33191" ...		
\$ Black or African American - female [State] 2023-24	: chr [1:57]	"115699" "1
" "31747" ...		
\$ White - male [State] 2023-24	: chr [1:57]	"197384" "3
" "194292" ...		
\$ White - female [State] 2023-24	: chr [1:57]	"184166" "2

```

" "184028" ...
$ Grade 12 Students - Black or African American [State] 2023-24      : chr [1:57] "15413" "24
" "5876" ...
$ Secondary Teachers [State] 2023-24                                : chr [1:57] "20188.71"
" "15996.72" ...
$ Elementary Teachers [State] 2023-24                                : chr [1:57] "18395.46"
" "30478.09" ...
$ Full-Time Equivalent (FTE) Staff [State] 2023-24                  : chr [1:57] "79963.90"
" "109199.43" ...
- attr(*, "spec")=
.. cols(
..   `State Name` = col_character(),
..   `State Abbr` = col_character(),
..   `ANSI/FIPS State Code [State] Latest available year` = col_character(),
..   `Total Number Operational School Districts [District] 2023-24` = col_double(),
..   `Total Number Operational Schools [Public School] 2023-24` = col_double(),
..   `Total Number Operational Charter Schools [Public School] 2023-24` = col_character(),
..   `Prekindergarten Students [State] 2023-24` = col_character(),
..   `Kindergarten Students [State] 2023-24` = col_character(),
..   `Grades 1-8 Students [State] 2023-24` = col_character(),
..   `Grades 9-12 Students [State] 2023-24` = col_character(),
..   `Grade 13 Students [State] 2023-24` = col_character(),
..   `Ungraded Students [State] 2023-24` = col_character(),
..   `American Indian/Alaska Native Students [State] 2023-24` = col_character(),
..   `Asian or Asian/Pacific Islander Students [State] 2023-24` = col_character(),
..   `Hispanic Students [State] 2023-24` = col_character(),
..   `Black or African American Students [State] 2023-24` = col_character(),
..   `White Students [State] 2023-24` = col_character(),
..   `Nat. Hawaiian or Other Pacific Isl. Students [State] 2023-24` = col_character(),
..   `Two or More Races Students [State] 2023-24` = col_character(),
..   `Total Enrollment (Exclude AE) for SY 2014-15 onward [State] 2023-24` = col_character(),
..   `Full-Time Equivalent (FTE) Teachers [State] 2023-24` = col_character(),
..   `Pupil/Teacher Ratio [State] 2023-24` = col_character(),
..   `State Name [State] 2023-24` = col_character(),
..   `Total Number of Districts with Enrollment [District] 2023-24` = col_double(),
..   `Total Number of School Districts [District] 2023-24` = col_double(),
..   `Total Number of Public Schools [Public School] 2023-24` = col_double(),
..   `Male Students [State] 2023-24` = col_character(),
..   `Female Students [State] 2023-24` = col_character(),
..   `Black or African American - male [State] 2023-24` = col_character(),
..   `Black or African American - female [State] 2023-24` = col_character(),
..   `White - male [State] 2023-24` = col_character(),
..   `White - female [State] 2023-24` = col_character(),
..   `Grade 12 Students - Black or African American [State] 2023-24` = col_character(),
..   `Secondary Teachers [State] 2023-24` = col_character(),
..   `Elementary Teachers [State] 2023-24` = col_character(),
..   `Full-Time Equivalent (FTE) Staff [State] 2023-24` = col_character()
.. )

```

```
- attr(*, "problems")=<externalptr>
```

0.3.3 Step 3: Clean the data

```
district_clean <- district %>%

# Make long variable names easier to use
clean_names() %>%

# Trim whitespace in all text fields
mutate(across(where(is.character), str_trim)) %>%

# Convert measure columns stored as character text to numeric.
# Keep state identifiers as character variables.
mutate(
  across(
    where(is.character) &
    !all_of(c(
      "state_name",
      "state_abbr",
      "state_name_state_2023_24"
    )),
    ~ parse_number(.x, na = c("", "NA", "-", "†"))
  )
) %>%

# Keep one state-name variable rather than two identical versions
select(-state_name_state_2023_24)

glimpse(district_clean)
```

Rows: 57

Columns: 35

\$ state_name	<chr> "ALABA~
\$ state_abbr	<chr> "AL", ~
\$ ansi_fips_state_code_state_latest_available_year	<dbl> 1, 2, ~
\$ total_number_operational_school_districts_district_2023_24	<dbl> 156, 5~
\$ total_number_operational_schools_public_school_2023_24	<dbl> 1524, ~
\$ total_number_operational_charter_schools_public_school_2023_24	<dbl> 18, 31~
\$ prekindergarten_students_state_2023_24	<dbl> 21924,~
\$ kindergarten_students_state_2023_24	<dbl> 55881,~
\$ grades_1_8_students_state_2023_24	<dbl> 447317~
\$ grades_9_12_students_state_2023_24	<dbl> 223528~
\$ grade_13_students_state_2023_24	<dbl> NA, NA~
\$ ungraded_students_state_2023_24	<dbl> NA, NA~

```

$ american_indian_alaska_native_students_state_2023_24 <dbl> 5997, ~
$ asian_or_asian_pacific_islander_students_state_2023_24 <dbl> 11016,~
$ hispanic_students_state_2023_24 <dbl> 84196,~
$ black_or_african_american_students_state_2023_24 <dbl> 236304~
$ white_students_state_2023_24 <dbl> 381550~
$ nat_hawaiian_or_other_pacific_isl_students_state_2023_24 <dbl> 831, 4~
$ two_or_more_races_students_state_2023_24 <dbl> 28756,~
$ total_enrollment_exclude_ae_for_sy_2014_15_onward_state_2023_24 <dbl> 748650~
$ full_time_equivalent_fte_teachers_state_2023_24 <dbl> 42857.~
$ pupil_teacher_ratio_state_2023_24 <dbl> 17.47,~
$ total_number_of_districts_with_enrollment_district_2023_24 <dbl> 150, 5~
$ total_number_of_school_districts_district_2023_24 <dbl> 159, 5~
$ total_number_of_public_schools_public_school_2023_24 <dbl> 1527, ~
$ male_students_state_2023_24 <dbl> 384500~
$ female_students_state_2023_24 <dbl> 364150~
$ black_or_african_american_male_state_2023_24 <dbl> 120605~
$ black_or_african_american_female_state_2023_24 <dbl> 115699~
$ white_male_state_2023_24 <dbl> 197384~
$ white_female_state_2023_24 <dbl> 184166~
$ grade_12_students_black_or_african_american_state_2023_24 <dbl> 15413,~
$ secondary_teachers_state_2023_24 <dbl> 20188.~
$ elementary_teachers_state_2023_24 <dbl> 18395.~
$ full_time_equivalent_fte_staff_state_2023_24 <dbl> 79963.~

```

0.3.4 Step 4: Feature engineering

We can create variables that answer educational questions.

```
str(district_clean)
```

```

tibble [57 x 35] (S3: tbl_df/tbl/data.frame)
 $ state_name           : chr [1:57] "ALABAMA" "ALASKA" "ALABAMA" "ALASKA"
 $ state_abbr           : chr [1:57] "AL" "AK" "AS" "AK"
 $ ansi_fips_state_code_state_latest_available_year : num [1:57] 1 2 60 4 5 59 6
 $ total_number_operational_school_districts_district_2023_24 : num [1:57] 156 54 1 685 30
 $ total_number_operational_schools_public_school_2023_24 : num [1:57] 1524 494 29 242
 $ total_number_operational_charter_schools_public_school_2023_24 : num [1:57] 18 31 NA 571 10
 $ prekindergarten_students_state_2023_24 : num [1:57] 21924 3320 NA 20
 $ kindergarten_students_state_2023_24 : num [1:57] 55881 9257 NA 7
 $ grades_1_8_students_state_2023_24 : num [1:57] 447317 78890 NA
 $ grades_9_12_students_state_2023_24 : num [1:57] 223528 39776 NA
 $ grade_13_students_state_2023_24 : num [1:57] NA NA NA NA NA
 $ ungraded_students_state_2023_24 : num [1:57] NA NA NA 533 17
 $ american_indian_alaska_native_students_state_2023_24 : num [1:57] 5997 28284 NA 4
 $ asian_or_asian_pacific_islander_students_state_2023_24 : num [1:57] 11016 6458 NA 3
 $ hispanic_students_state_2023_24 : num [1:57] 84196 10076 NA 1

```

```

$ black_or_african_american_students_state_2023_24      : num [1:57] 236304 2955 NA 6
$ white_students_state_2023_24                          : num [1:57] 381550 61502 NA
$ nat_hawaiian_or_other_pacific_isl_students_state_2023_24 : num [1:57] 831 4148 NA 401
$ two_or_more_races_students_state_2023_24             : num [1:57] 28756 17820 NA 4
$ total_enrollment_exclude_ae_for_sy_2014_15_onward_state_2023_24 : num [1:57] 748650 131243 NA
$ full_time_equivalent_fte_teachers_state_2023_24      : num [1:57] 42858 7221 NA 4
$ pupil_teacher_ratio_state_2023_24                    : num [1:57] 17.5 18.2 NA 22
$ total_number_of_districts_with_enrollment_district_2023_24 : num [1:57] 150 54 0 664 26
$ total_number_of_school_districts_district_2023_24    : num [1:57] 159 54 1 722 30
$ total_number_of_public_schools_public_school_2023_24 : num [1:57] 1527 498 29 255
$ male_students_state_2023_24                          : num [1:57] 384500 67646 NA
$ female_students_state_2023_24                        : num [1:57] 364150 63597 NA
$ black_or_african_american_male_state_2023_24         : num [1:57] 120605 1488 NA 3
$ black_or_african_american_female_state_2023_24      : num [1:57] 115699 1467 NA 3
$ white_male_state_2023_24                             : num [1:57] 197384 31786 NA
$ white_female_state_2023_24                           : num [1:57] 184166 29716 NA
$ grade_12_students_black_or_african_american_state_2023_24 : num [1:57] 15413 241 NA 58
$ secondary_teachers_state_2023_24                     : num [1:57] 20189 3673 NA 1
$ elementary_teachers_state_2023_24                    : num [1:57] 18395 2985 NA 3
$ full_time_equivalent_fte_staff_state_2023_24         : num [1:57] 79964 16423 NA
..- attr(*, "problems")= tibble [1 x 4] (S3: tbl_df/tbl/data.frame)
.. ..$ row      : int 51
.. ..$ col      : int NA
.. ..$ expected: chr "a number"
.. ..$ actual   : chr "‡"

```

```

district_features <- district_clean %>%
  mutate(
    enrollment_per_operational_district =
      if_else(
        total_number_operational_school_districts_district_2023_24 > 0,
        total_enrollment_exclude_ae_for_sy_2014_15_onward_state_2023_24 /
          total_number_operational_school_districts_district_2023_24,
        NA_real_
      ),

    schools_per_operational_district =
      if_else(
        total_number_operational_school_districts_district_2023_24 > 0,
        total_number_operational_schools_public_school_2023_24 /
          total_number_operational_school_districts_district_2023_24,
        NA_real_
      ),

    charter_school_share =
      if_else(
        total_number_operational_schools_public_school_2023_24 > 0,

```

```

        total_number_operational_charter_schools_public_school_2023_24 /
        total_number_operational_schools_public_school_2023_24,
        NA_real_
    ),

    elementary_teacher_share =
    if_else(
        full_time_equivalent_fte_teachers_state_2023_24 > 0,
        elementary_teachers_state_2023_24 /
        full_time_equivalent_fte_teachers_state_2023_24,
        NA_real_
    )
)

district_features %>%
  select(
    state_name,
    total_enrollment_exclude_ae_for_sy_2014_15_onward_state_2023_24,
    pupil_teacher_ratio_state_2023_24,
    enrollment_per_operational_district,
    schools_per_operational_district,
    charter_school_share,
    elementary_teacher_share
  ) %>%
  glimpse()

```

Rows: 57

Columns: 7

```

$ state_name                                <chr> "ALABA~
$ total_enrollment_exclude_ae_for_sy_2014_15_onward_state_2023_24 <dbl> 748650~
$ pupil_teacher_ratio_state_2023_24         <dbl> 17.47,~
$ enrollment_per_operational_district        <dbl> 4799.0~
$ schools_per_operational_district           <dbl> 9.7692~
$ charter_school_share                      <dbl> 0.0118~
$ elementary_teacher_share                  <dbl> 0.4292~

```

What kinds of questions might you answer with this data set?

0.4 Recommended readings

Knaflig, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals*. Wiley.